

Unit 6 Progress Check: MCQ

1. $\text{HCl}(aq) + \text{NH}_3(aq) \rightarrow \text{NH}_4^+(aq) + \text{Cl}^-(aq)$
- The chemical reaction between $\text{HCl}(aq)$ and $\text{NH}_3(aq)$ is represented above. A student combines equimolar amounts of $\text{HCl}(aq)$ and $\text{NH}_3(aq)$, both solutions initially at 24°C , in a coffee-cup calorimeter. The student observes that the mixture reaches a temperature of 28°C . Based on the experimental results, which of the following can be concluded about the reaction?
- (A) It is an endothermic process, because energy is released by the reaction and is gained by the reaction mixture.
- (B) It is an endothermic process, because energy is absorbed by the reaction and is lost from the reaction mixture.
- (C) It is an exothermic process, because energy is released by the reaction and is gained by the reaction mixture.
- (D) It is an exothermic process, because energy is absorbed by the reaction and is lost from the reaction mixture.
2. When pellets of $\text{NaOH}(s)$ are added to a flask of water, it is observed that the temperature of the water increases as the pellets dissolve. Which of the following claims about the observed dissolution of $\text{NaOH}(s)$ in water is most accurate?
- (A) It is an endothermic process because heat energy is absorbed by the water as the $\text{NaOH}(s)$ dissolves in it.
- (B) It is an endothermic process because heat energy is released by the water as the $\text{NaOH}(s)$ dissolves in it.
- (C) It is an exothermic process because heat energy is absorbed by the water as the $\text{NaOH}(s)$ dissolves in it.
- (D) It is an exothermic process because heat energy is released by the water as the $\text{NaOH}(s)$ dissolves in it.
3. An advertisement for a commercial hand warmer claims that the hand warmer works because a chemical reaction in the hand warmer draws out the body's own natural heat, causing a warming effect. Which of the following states the accuracy of the claim in the advertisement and best provides a correct scientific justification of the claim?



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- (A) The advertisement's claim is inaccurate because heat flowing from the hands to the warmer would only happen if the chemical reaction was endothermic, which would cause the hands to feel colder.
- (B) The advertisement's claim is inaccurate because heat flowing from the hands to the warmer would only happen if the chemical reaction was exothermic, which would cause the hands to feel colder.
- (C) The advertisement's claim is accurate because heat flowing from the hands to the warmer would only happen if the chemical reaction was endothermic, which would cause the hands to feel warmer.
- (D) The advertisement's claim is accurate because heat flowing from the hands to the warmer would only happen if the chemical reaction was exothermic, which would cause the hands to feel warmer.
4. A student carefully drops a **9.0 g** solid **Zn** pellet initially at **50.0°C** into an insulated cup containing **30.0 g** of water at **27.8°C**. The student predicts that the temperature of the water will increase after the pellet is added. Which of the following statements is the best justification for the student's prediction?
- (A) The metallic bonds between **Zn** atoms will break when the **Zn** is exposed to the water molecules, releasing energy that will be absorbed by the water molecules.
- (B) Collisions between the water molecules and the surface of the **Zn** pellet will result in the transfer of energy, increasing the average kinetic energy of the water molecules.
- (C) The strength of the hydrogen bonds between the water molecules will increase when the **Zn** pellet is added, decreasing the average kinetic energy of the water molecules.
- (D) Collisions between **Zn** atoms in the solid will increase in frequency when the **Zn** is exposed to the water molecules, resulting in the transfer of energy to the surroundings.
5. Two aqueous **NaCl** solutions of equal volume and concentration were kept in flasks and held at different temperatures. The two solutions were combined in a larger flask. Based on this information, which of the following predictions is correct?



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- (A) The average kinetic energy of the particles in the cooler solution will decrease as they collide with the particles from the warmer solution.
- (B) The average kinetic energy of the particles in the cooler solution will increase as they collide with the particles from the warmer solution.
- (C) The particles from the cooler solution and the particles from the warmer solution will reach thermal equilibrium through collisions, at which time the average kinetic energy of the particles in the mixture will be lower than the average kinetic energy that the particles had in the cooler solution.
- (D) The particles from the cooler solution and the particles from the warmer solution will reach thermal equilibrium through collisions, at which time the average kinetic energy of the particles in the mixture will be higher than the average kinetic energy that the particles had in the warmer solution.

6.

Sample	Initial Volume (mL)	Initial Temperature (°C)
1	125.0	36.0
2	250.0	18.0

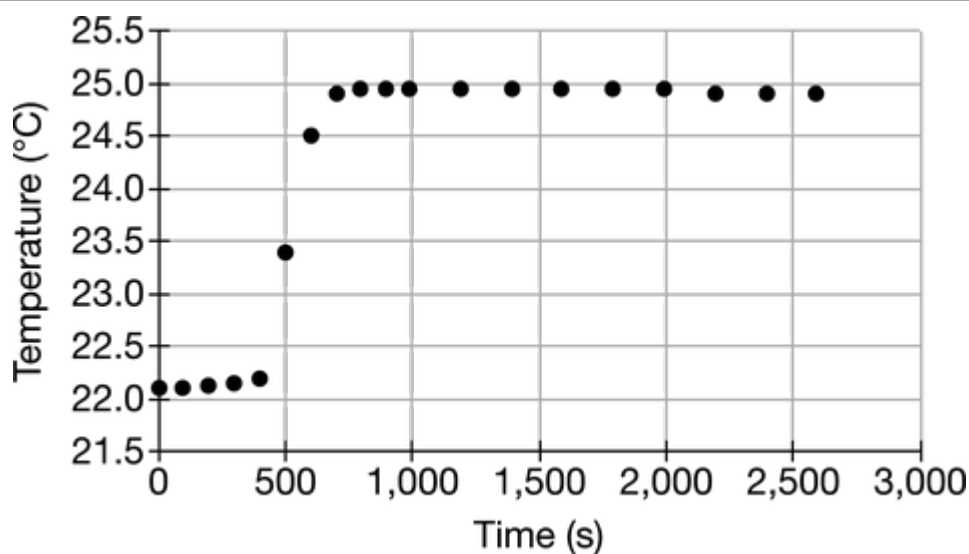
The table provides data for two $\text{CH}_3\text{OH}(l)$ samples. Based on this information, which of the following statements describes what happens when these samples are initially mixed, and why?

- (A) Thermal energy is transferred between the CH_3OH molecules in sample 1 and the CH_3OH molecules in sample 2 because both samples contain the same substance.
- (B) The CH_3OH molecules from sample 2 transfer thermal energy to the CH_3OH molecules from sample 1 through collisions because there are more moles of molecules in sample 2.
- (C) The CH_3OH molecules from sample 1 transfer thermal energy to the CH_3OH molecules from sample 2 through collisions because sample 1 has a higher density.
- (D) The CH_3OH molecules from sample 1 transfer thermal energy to the CH_3OH molecules from sample 2 through collisions because the average kinetic energy of the molecules in sample 1 is greater.



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7.



The graph above shows the changes in temperature recorded for the **2.00 L** of **H₂O** surrounding a constant-volume container in which a **1.00 g** sample of benzoic acid was combusted. Assume that heat was not absorbed by the container or lost to the surroundings, that the density of **H₂O** is **1.00 g/mL**, and that the specific heat capacity of **H₂O** is about **4.2 J/g · °C**. Based on this information, estimate how much heat was released from the combustion of the benzoic acid sample.

- (A) 0.013 kJ
- (B) 25 kJ
- (C) 180 kJ
- (D) 210 kJ



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8.	Substance	Mass (g)	Initial Temperature (°C)	ΔT (°C)	Specific Heat Capacity (J/g · °C)	q
	Al	20.0	90.	−5.9	0.90	?
	H ₂ O	40.0	25	6	4.2	?

For an experiment, a **20.0 g** piece of **Al** was placed inside an insulated cup filled with **40.0 g** of **H₂O**. The data are given in the table above. Assuming no heat is absorbed by the container or the surroundings, which of the following equations provides the correct mathematical relationship for the transfer of heat q between **Al** and **H₂O**?

(A) $-q_{\text{Al}} = 12 \times q_{\text{H}_2\text{O}}$

(B) $-q_{\text{Al}} = 6 \times q_{\text{H}_2\text{O}}$

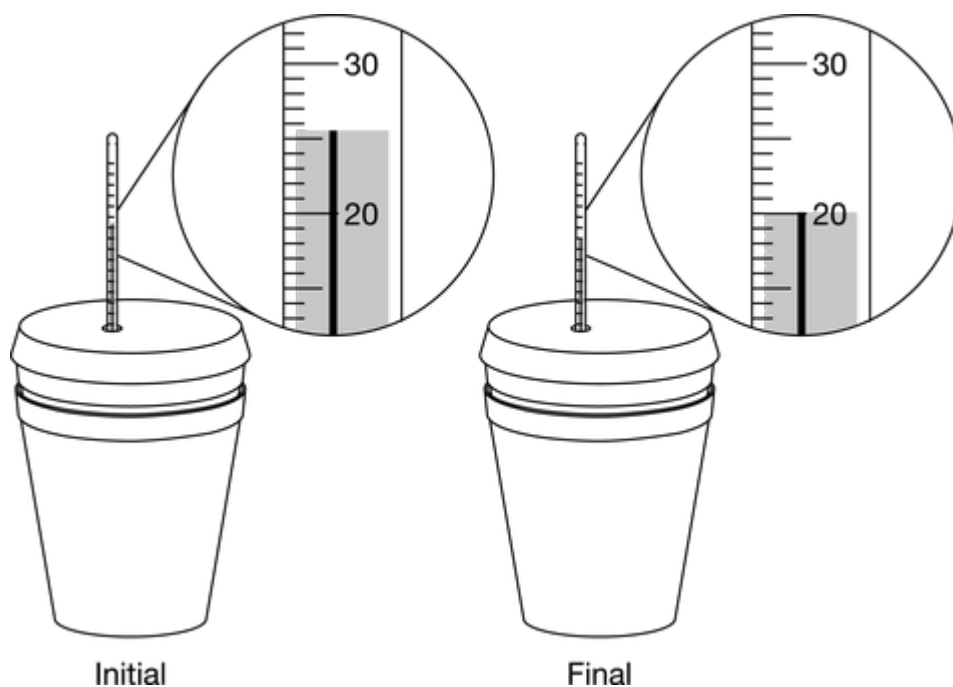
(C) $-q_{\text{Al}} = 2 \times q_{\text{H}_2\text{O}}$

(D) $-q_{\text{Al}} = q_{\text{H}_2\text{O}}$



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9.



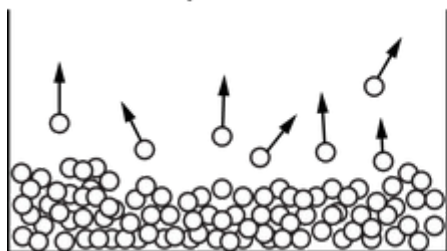
A student dissolves 8.00 g of $\text{NH}_4\text{NO}_3(s)$ in $100.\text{ g}$ of $\text{H}_2\text{O}(l)$ in a coffee-cup calorimeter. Based on the initial and final temperatures of the mixture in degrees Celsius shown in the diagram above the laboratory setup, what is the calculated ΔT of the water reported to the appropriate number of significant figures?

- (A) -5.500°C
- (B) -5.50°C
- (C) -5.5°C
- (D) -6°C

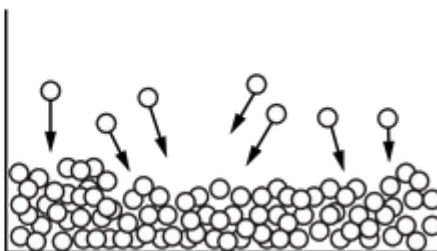


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10. Evaporation



Condensation



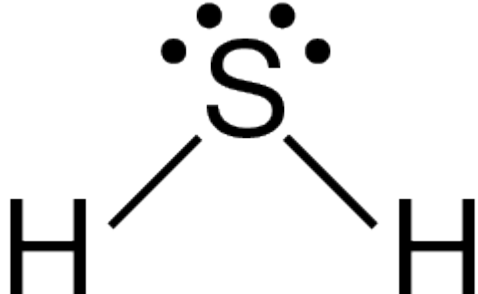
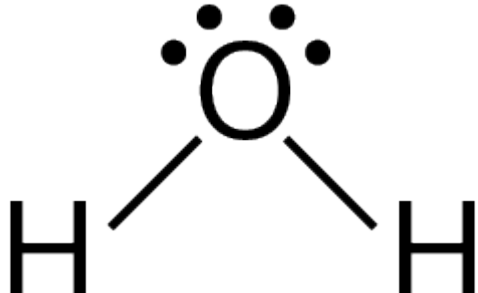
Particulate models of the evaporation of liquid water and the condensation of water vapor are shown above. Based on these models, which of the following accurately compares the energy changes associated with each of the phase changes?

- (A) The amount of energy absorbed by one mole of water molecules as they escape the liquid is greater than the amount of energy released by one mole of water molecules as they come together to form a liquid.
- (B) The amount of energy absorbed by one mole of water molecules as they escape the liquid is less than the amount of energy released by one mole of water molecules as they come together to form a liquid.
- (C) The amount of energy absorbed by one mole of water molecules as they escape the liquid is equal to the amount of energy released by one mole of water molecules as they come together to form a liquid.
- (D) The amount of energy absorbed by one mole of water molecules escaping the liquid can be either greater than or less than the amount of energy released by one mole of water molecules as they come together.



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11.

Substance	ΔH_{vap} (kJ/mol)
	20
	40

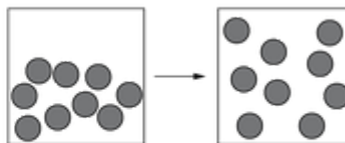
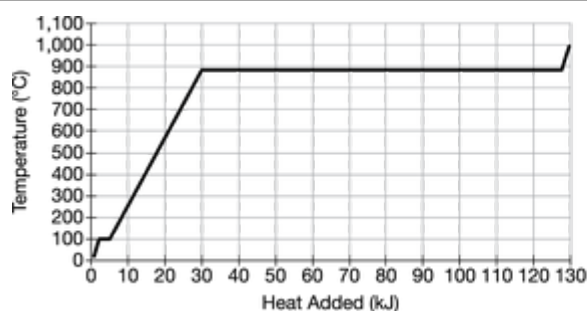
The table above provides the values for some physical properties of H_2S and H_2O . Approximately, how many moles of H_2S must be condensed to release as much heat as would be released when 1 mole of H_2O is condensed?

- (A) 0.5 mole of H_2S
- (B) 1 mole of H_2S
- (C) 2 moles of H_2S
- (D) 4 moles of H_2S



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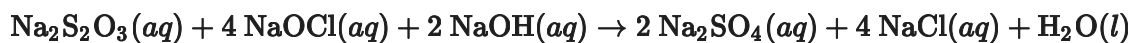
12.



The heating curve for 1.0 mole of **Na**, initially at **25.0°C**, is shown above at the left. Which of the following best explains the change in heat when 0.50 mole of **Na** undergoes the transition shown in the diagram above to the right?

- (A) Approximately **13 kJ** of heat are absorbed as a result of the increase in potential energy between the **Na** atoms.
- (B) Approximately **49 kJ** of heat are absorbed to overcome the attractive forces acting between **Na** atoms.
- (C) Approximately **98 kJ** of heat are released as a result of the decrease in the kinetic energy of the **Na** atoms.
- (D) Approximately **120 kJ** of heat are released to decrease the potential energy between **Na** atoms.

13.

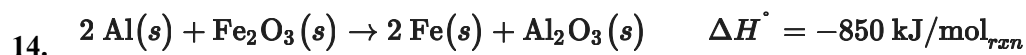


The enthalpy change for the reaction represented by the chemical equation shown above is $\Delta H^\circ = -1236 \text{ kJ/mol}_{\text{rxn}}$. When **2.00 mol** of **NaOCl(aq)** reacts completely with an excess of **Na₂S₂O₃(aq)** and of **NaOH(aq)**, which of the following enthalpy changes occurs?

- (A) **618 kJ** of heat is absorbed.
- (B) **2470 kJ** of heat is absorbed.
- (C) **618 kJ** of heat is released.
- (D) **2470 kJ** of heat is released.

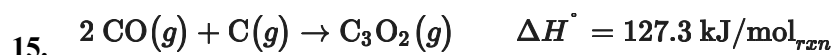


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The chemical equation shown above represents the thermite reaction. What is the approximate amount of heat released when 108 g of $\text{Al}(s)$ reacts with excess $\text{Fe}_2\text{O}_3(s)$?

- (A) 210 kJ
- (B) 430 kJ
- (C) 850 kJ
- (D) 1700 kJ



The equation shown above represents an endothermic reaction between $\text{CO}(g)$ and $\text{C}(g)$. What is the amount of heat absorbed when 1.00 mol of $\text{CO}(g)$ reacts with an excess $\text{C}(g)$?

- (A) 31.8 kJ
- (B) 63.7 kJ
- (C) 127.3 kJ
- (D) 254.6 kJ

16.

Type of Bond	Average Bond Enthalpy (kJ/mol)
O – O	150
O = O	500

$2 \text{O}_3(g) \rightarrow 3 \text{O}_2(g) \quad \Delta H^\circ_{\text{rxn}} = -300 \text{ kJ/mol}_{\text{rxn}}$ The conversion of ozone to diatomic oxygen is represented by the equation above. Based on the data in the table above, what is the approximate average bond enthalpy for the oxygen-to-oxygen bonds in ozone?



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- (A) 0 kJ/mol
- (B) 150 kJ/mol
- (C) 300 kJ/mol
- (D) 500 kJ/mol

17.

Bond Type	Average Bond Enthalpy (kJ/mol)
C – H	414
C – O	360
C = O	799
O – H	464

The combustion of methane proceeds according to the chemical equation $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$. The table above provides the average bond enthalpy for selected bonds. The calculated enthalpy change for the reaction, ΔH_{rxn} , is -802 kJ/mol . Which of the following provides the mathematical procedure to estimate the bond enthalpy per mole of O_2 ?

- (A) $\frac{1}{2}[-802 + 2(799) + 4(464) - 4(414)] \text{ kJ/mol}$
- (B) $2[-802 + 799 + 2(464) - 2(414)] \text{ kJ/mol}$
- (C) $\frac{1}{2}[-802 - 2(360) + 4(464) + 4(414)] \text{ kJ/mol}$
- (D) $2[-802 - 360 + 2(464) + 414] \text{ kJ/mol}$



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18.

Bond Type	Average Bond Enthalpy
H – H	436 kJ/mol
Cl – Cl	243 kJ/mol
H – Cl	432 kJ/mol

The formation of hydrogen chloride gas is represented by the chemical equation $\text{H}_2(g) + \text{Cl}_2(g) \rightarrow 2 \text{HCl}(g)$. Based on the bond enthalpy data in the table above, what is the approximate enthalpy change for the reaction?

- (A) $-247 \text{ kJ/mol}_{\text{rxn}}$
- (B) $-185 \text{ kJ/mol}_{\text{rxn}}$
- (C) $+185 \text{ kJ/mol}_{\text{rxn}}$
- (D) $+247 \text{ kJ/mol}_{\text{rxn}}$

19.

Substance	ΔH_f° (kJ/mol)
$\text{CaCO}_3(s)$	-1210
$\text{CaO}(s)$	-640
$\text{CO}_2(g)$	-390

CaCO_3 decomposes according to the balanced equation $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$. Based on the standard enthalpies of formation provided in the table above, what is the approximate enthalpy change of the reaction?

- (A) -180 kJ
- (B) $+180 \text{ kJ}$
- (C) -1460 kJ
- (D) $+1460 \text{ kJ}$



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20.

Substance	ΔH_f° (kJ/mol)
$\text{SO}_2(g)$	-296.1
$\text{SO}_3(g)$	-395.2
$\text{O}_2(g)$	0.0

Based on the information above, what is ΔH° for the reaction $\text{SO}_2(g) + \frac{1}{2} \text{O}_2(g) \rightarrow \text{SO}_3(g)$?

- (A) $-591.3 \text{ kJ/mol}_{\text{rxn}}$
- (B) $-99.1 \text{ kJ/mol}_{\text{rxn}}$
- (C) $+99.1 \text{ kJ/mol}_{\text{rxn}}$
- (D) $+591.3 \text{ kJ/mol}_{\text{rxn}}$

21.

Substance	ΔH_f° (kJ/mol)
$\text{CH}_4(g)$	-75
$\text{O}_2(g)$	0
$\text{CO}_2(g)$	-390
$\text{H}_2\text{O}(g)$	-240

Natural gas consists primarily of CH_4 , which is combusted according to the following chemical equation.

$$\text{CH}_4(g) + 2 \text{O}_2(g) \rightarrow \text{CO}_2(g) + 2 \text{H}_2\text{O}(g)$$

Based on the standard enthalpies of formation in the table above, which of the following expressions give the approximate enthalpy change for the reaction ($\Delta H_{\text{rxn}}^\circ$) ?



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- (A) $\Delta H_{rxn}^{\circ} = (-75 \text{ kJ/mol}) - [(-390 \text{ kJ/mol}) + 2(-240 \text{ kJ/mol})]$
- (B) $\Delta H_{rxn}^{\circ} = (-75 \text{ kJ/mol}) - [(-390 \text{ kJ/mol}) + (-240 \text{ kJ/mol})]$
- (C) $\Delta H_{rxn}^{\circ} = [(-390 \text{ kJ/mol}) + 2(-240 \text{ kJ/mol})] - (-75 \text{ kJ/mol})$
- (D) $\Delta H_{rxn}^{\circ} = [(-390 \text{ kJ/mol}) + (-240 \text{ kJ/mol})] - (-75 \text{ kJ/mol})$

22.

$\text{C}_2\text{H}_6(g) + \frac{7}{2} \text{O}_2(g) \rightarrow 2 \text{CO}_2(g) + 3 \text{H}_2\text{O}(l)$	$\Delta H^{\circ} = -1560 \text{ kJ}$
$\text{C}(\text{graphite}) + \text{O}_2(g) \rightarrow \text{CO}_2(g)$	$\Delta H^{\circ} = -394 \text{ kJ}$

Given the equations and the values of ΔH° for the combustion of $\text{C}_2\text{H}_6(g)$ and the combustion of $\text{C}(\text{graphite})$ represented above, which of the following additional information is needed to determine the values of ΔH° for the overall reaction $2 \text{C}(\text{graphite}) + 3 \text{H}_2(g) \rightarrow \text{C}_2\text{H}_6(g)$?

- (A) $\text{C}(\text{diamond}) \rightarrow \text{C}(\text{graphite}) \quad \Delta H^{\circ} = -2 \text{ kJ}$
- (B) $\text{C}_2\text{H}_4(g) + \text{H}_2(g) \rightarrow \text{C}_2\text{H}_6(g) \quad \Delta H^{\circ} = -134 \text{ kJ}$
- (C) $\text{H}_2(g) + \frac{1}{2} \text{O}_2(g) \rightarrow \text{H}_2\text{O}(l) \quad \Delta H^{\circ} = -286 \text{ kJ}$
- (D) $\text{H}_2\text{O}(l) \rightarrow \text{H}_2\text{O}(g) \quad \Delta H^{\circ} = +41 \text{ kJ}$

23.

Reaction 1:	$2 \text{O}_3(g) \rightarrow 3 \text{O}_2(g)$
Reaction 2:	?
Overall reaction:	$2 \text{O}_3(g) + 2 \text{NO}(g) \rightarrow 2 \text{NO}_2(g) + 2 \text{O}_2(g)$

The enthalpy of the overall reaction represented above can be determined by adding the enthalpies of reactions 1 and 2. Which of the following could be reaction 2?



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- (A) $\text{NO}(g) + \text{O}(g) \rightarrow \text{NO}_2(g)$
- (B) $(\text{NO})_2(g) \rightarrow 2 \text{NO}(g)$
- (C) $2 \text{NO}(g) + \text{O}_2(g) \rightarrow 2 \text{NO}_2(g)$
- (D) $2 \text{NO}_2(g) + \text{O}_3(g) \rightarrow \text{N}_2\text{O}_5(g) + \text{O}_2(g)$

24.

Chemical Equation	ΔH° (kJ/mol _{rxn})
$\text{H}_2(g) + \frac{1}{2} \text{O}_2(g) \rightarrow \text{H}_2\text{O}(l)$	−286
$\text{N}_2(g) + 3 \text{O}_2(g) + \text{H}_2(g) \rightarrow 2 \text{HNO}_3(l)$	−348

The table above provides the values for the standard enthalpy change for two chemical reactions. A student needs to calculate the enthalpy change for the reaction $2 \text{HNO}_3(l) \rightarrow \text{N}_2\text{O}_5(g) + \text{H}_2\text{O}(l)$ under standard conditions using Hess's law. In addition to the values given in the table, what other information will the student need to perform the calculation?

- (A) ΔH_f° for $\text{N}_2(l)$
- (B) ΔH_f° for $\text{N}_2\text{O}_5(g)$
- (C) ΔH_f° for $\text{H}_2\text{O}(g)$
- (D) ΔH_f° for $\text{H}_2(l)$